Master Formula Inventory

This file contains all formulas physically extracted from source documents.

From Source: `formulas.typ`

Variable: `flat_sums_btherom`

Display:

$$x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\sum_{d=0}^{c}\left[\binom{c}{d}\ldots\left[\sum_{n_l=0}^{n_{l-1}}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]a^{c-d}\right]a^{b-c}\right]a^{r-b}\right]}^{l \text{ sums}}$$

Variable: `falling_sums_btherom`

Display: for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid d$ in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right) \Rightarrow$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\ldots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}a^{n_{l-1}-n_l}(x-la)^{n_l}\right]\right]\ldots\right]\right]\right]}^{l \text{ sums}}$$

$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[\sum_{c=0}^{b-1}\left[\binom{b}{c}a^{b-c}\ldots\left[\sum_{n_{l-1}=0}^{n_{l-2}-1}\left[\binom{n_{l-2}}{n_{l-1}}a^{n_{l-2}-n_{l-1}}(x-la)^{n_{l-1}}\right]\right]\ldots\right]\right]\right]}^{l-1 \text{ sums}}$$

$$+\ldots+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}a^{r-b}\left[(x-2a)^b\right]\right]}^{1 \text{ sums}}$$

$$+\overbrace{(x-a)^r}^{0 \text{ sums}}$$

Variable: `falling_sum_lr_btherom`

Display:

$$\text{for all } a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$$

$$x^r = \sum_{b=0}^{l-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}(x-(b+1)a)^c a^{r-c}\right]\right] + (x-la)^r$$

Variable: `falling_sum_lr_btherom_reversed`

Display:

$$\text{for all } a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$$

$$x^r = \sum_{b=(l-1)}^{x-a}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}(b)^c a^{r-c}\right]\right] + (x-la)^r$$

Variable: `n_falling_sums_sub_freedom_btherom`

Display:

$$\text{for all } r \in \mathbb{Z}^+ \mid n_0 \in R \mid n_0 \neq 0 \mid n_1 ... n_n \in \mathbb{Z}^+$$

$$x^r = \overbrace{\sum_{b_n=0}^{n_n-1}\left[...\sum_{b_2=0}^{n_2-1}\left[\sum_{b_1=0}^{n_1-1}\left[\sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}((...(x-b_n n_{n-1}...n_2 n_1 n_0)... - b_2 n_1 n_0) - (b_1+1)n_0)^{b_0} n_0^{r-b_0}\right]\right]\right]}^{n \text{ sums}} + (x - n_n ... n_2 n_1 n_0)^r$$

Variable: `n_falling_sums_add_freedom_btherom`

Display:

$$\text{for all } r \in \mathbb{Z}^+ \mid n_0 \in R \mid n_0 \neq 0 \mid n_1 ... n_n \in \mathbb{Z}^+$$

$$x^r = -1\overbrace{\sum_{b_n=0}^{n_n-1}\left[...\sum_{b_2=0}^{n_2-1}\left[\sum_{b_1=0}^{n_1-1}\left[\sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}(((x + b_n n_n ... n_1 n_0) + ... + b_2 n_1 n_0) + b_1 n_0)^{b_0} n_0^{r-b_0}\right]\right]\right]}^{n \text{ sums}} + ((x + n_n ... n_2 n_1 n_0)^r$$

Variable: `binom_hockey`

Display:

$$\binom{a}{n} = \sum_{b=0}^{a-1}\left[\binom{b}{n-1}\right] + \binom{0}{n}$$

Variable: `binom_hockey_recurssion`

Display:

$$\text{for all } n \in \mathbb{Z}_{\geq} 0$$

$$\binom{a}{n} = \overbrace{\sum_{b=0}^{a-1}\left[\sum_{c=0}^{b-1}\left[...\sum_{n_n=0}^{n_{n-1}-1}[1]...\right]\right]}^{n \text{ sums}} + \binom{0}{n}$$